



COMPUTER ORGANIZATION

Topic: Bus Structure For NIMCET [Date: 29.04.2026]



01. A system has a 20-bit address bus and 16-bit data bus. If memory is byte-addressable, max memory is:
A) 1 MB B) 2 MB
C) 16 MB D) 64 KB
02. If system is word-addressable (word = 16 bits) with 20-bit address bus, memory capacity is:
A) 1 MB B) 2 MB
C) 4 MB D) 512 KB
03. Increasing data bus width from 16 → 32 bits:
A) Doubles memory
B) Doubles bandwidth
C) Doubles address space
D) No effect
04. If bus frequency = 100 MHz and width = 32 bits, bandwidth:
A) 3200 Mbps B) 400 MB/s
C) 3200 MB/s D) 3.2 GB/s
05. Address bus size required for 4 GB memory:
A) 30 bits B) 32 bits
C) 34 bits D) 36 bits
06. If address bus = 12 bits and memory is word-addressable (word=4 bytes):
A) 4 KB B) 16 KB
C) 8 KB D) 2 KB
07. Data bus width affects:
A) Latency B) Throughput
C) Addressing D) Control
08. If data bus = 8 bits and instruction = 32 bits, fetch requires:
A) 1 cycle B) 2 cycles
C) 4 cycles D) 8 cycles
09. Address bus = n , memory = 1 TB → n ?
A) 30 B) 32
C) 40 D) 36
10. Control bus includes bidirectional signals because:
A) Data flows both ways
B) CPU receives interrupts
C) Memory sends addresses
D) None
11. If address bus = 10, data bus = 16, memory size:
A) 1 KB B) 2 KB
C) 1024 words D) Both B & C
12. Increasing address bus increases:
A) Bandwidth B) Capacity
C) Speed D) Cache
13. Bus width = 64 bits → data per cycle:
A) 8 bytes B) 64 bytes
C) 4 bytes D) 16 bytes
14. If bus speed doubles, bandwidth:
A) Doubles B) Halves
C) Same D) Depends
15. Address bus determines:
A) Maximum memory B) Data transfer
C) Speed D) Cache
16. Which bus uses tri-state logic?
A) Data B) Address
C) Control D) All
17. Bus contention occurs when:
A) Multiple devices read
B) Multiple devices write simultaneously
C) CPU idle
D) Memory idle
18. Which signal is NOT from control bus?
A) RD B) WR
C) Address lines D) Clock
19. Synchronous bus uses:
A) Clock B) Handshake
C) Interrupt D) Cache
20. Asynchronous bus uses:
A) Clock B) Handshake signals
C) Cache D) Interrupt
21. Which affects memory addressing?
A) Data bus B) Address bus
C) Control bus D) Clock
22. Data bus carries:
A) Address
B) Data & instructions
C) Signals
D) Clock

23. **Address bus is unidirectional because:**
 A) Memory sends data
 B) CPU sends address
 C) Data flows both ways
 D) None
24. **Control bus direction:**
 A) Always unidirectional
 B) Always bidirectional
 C) Mixed
 D) None
25. **Bus arbitration ensures:**
 A) Speed
 B) Fair access
 C) Memory
 D) Cache
26. **Daisy chaining arbitration is:**
 A) Centralized
 B) Distributed
 C) Hybrid
 D) None
27. **Which bus defines instruction fetch cycles?**
 A) Data
 B) Address
 C) Control
 D) Cache
28. **Interrupt signal travels via:**
 A) Data
 B) Address
 C) Control
 D) System
29. **System bus excludes:**
 A) Data
 B) Address
 C) Control
 D) ALU
30. **Bus speed measured in:**
 A) Bytes
 B) Hz
 C) Bits
 D) Volts
31. **If data bus increases but address bus same:**
 A) More memory
 B) Faster transfer
 C) More addressing
 D) No change
32. **Address bus increase does NOT affect:**
 A) Capacity
 B) Speed
 C) Address space
 D) Memory size
33. **Bus bandwidth depends on:**
 A) Width
 B) Frequency
 C) Both
 D) None
34. **Control bus does NOT carry:**
 A) Clock
 B) Interrupt
 C) Data
 D) Read
35. **If instruction size > data bus:**
 A) Single fetch
 B) Multiple fetch
 C) Error
 D) Cache
36. **Memory capacity = $2^n \times \text{word size}$ applies when:**
 A) Byte-addressable
 B) Word-addressable
 C) Cache
 D) None
37. **Which improves throughput?**
 A) Increase data bus
 B) Increase address bus
 C) Decrease speed
 D) None
38. **Which bus handles synchronization?**
 A) Data
 B) Address
 C) Control
 D) None
39. **Bus request/grant used in:**
 A) Addressing
 B) Arbitration
 C) Data transfer
 D) Cache
40. **Which is fastest?**
 A) Narrow bus
 B) Wide bus
 C) Same
 D) None
41. **Address bus determines:**
 A) Instruction size
 B) Memory capacity
 C) Speed
 D) Data
42. **Control bus ensures:**
 A) Timing
 B) Signals
 C) Coordination
 D) All
43. **Data bus determines:**
 A) Transfer size
 B) Memory
 C) Address
 D) Cache
44. **Bus is shared resource → leads to:**
 A) Conflict
 B) Contention
 C) Arbitration
 D) All
45. **Address bus is:**
 A) Bidirectional
 B) Unidirectional
 C) Mixed
 D) None
46. **System bus connects:**
 A) CPU
 B) Memory
 C) I/O
 D) All
47. **Increasing bus frequency:**
 A) Increases bandwidth
 B) Increases memory
 C) No effect
 D) Decreases speed
48. **Which affects latency?**
 A) Bus width
 B) Bus speed
 C) Both
 D) None
49. **Bus arbitration avoids:**
 A) Deadlock
 B) Conflict
 C) Cache miss
 D) Page fault
50. **Control bus signals include:**
 A) Read/Write
 B) Interrupt
 C) Clock
 D) All